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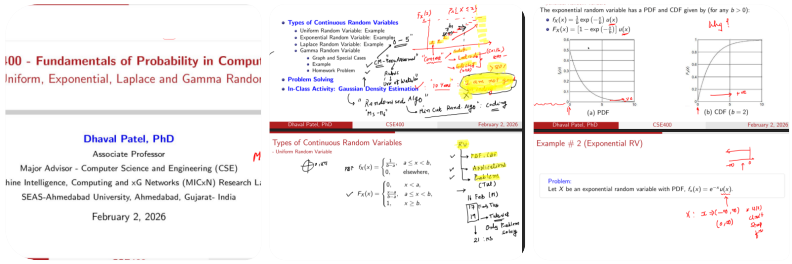

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CSE400 – Fundamentals of Probability in Computing

Lecture 9 Scribe

Uniform, Exponential, Laplace and Gamma Random Variables

Instructor: Dhaval Patel, PhD

Date: February 2, 2026

1. Introduction and Lecture Scope

This lecture introduces **types of continuous random variables** and discusses their mathematical formulation, properties, and applications. The lecture specifically covers:

1. Uniform Random Variable
2. Exponential Random Variable
3. Laplace Random Variable
4. Gamma Random Variable
5. Graphical and example-based understanding
6. Problem solving and Gaussian density estimation discussion

The material establishes the mathematical definitions of distributions using **Probability Density Function (PDF)** and **Cumulative Distribution Function (CDF)**.

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2. Types of Continuous Random Variables

Continuous random variables are defined using probability density functions instead of probability mass functions. The lecture focuses on commonly used continuous distributions used in computing and modelling scenarios.

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3. Uniform Random Variable

3.1 Definition

A random variable X is called a **Uniform Random Variable** over the interval $[a, b]$ if the probability density function is constant over this interval and zero elsewhere.

3.2 Probability Density Function (PDF)

The PDF of a uniform random variable is given by:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x < b \\ 0, & \text{elsewhere} \end{cases}$$

Explanation

1. The probability is evenly distributed across the interval $[a, b]$.

2. Since probability density must integrate to 1:

$$\int_a^b f_X(x)dx = 1$$

3. Because the density is constant over the interval:

$$f_X(x) = \frac{1}{b-a}$$

3.3 Cumulative Distribution Function (CDF)

The CDF is obtained by integrating the PDF.

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$$

3.4 Step-by-Step Derivation of CDF

Case 1: $x < a$

Since the distribution starts at a,

$$F_X(x) = 0$$

Case 2: $a \leq x < b$

$$F_X(x) = \int_a^x \frac{1}{b-a} dt$$

Evaluating integral:

$$F_X(x) = \frac{x-a}{b-a}$$

Case 3: $x \geq b$

Entire probability mass is accumulated:

$$F_X(x) = 1$$

3.5 Observations from Lecture Discussion

The lecture emphasizes:

- Uniform distribution is often treated as a **reference distribution**.
- It is used in modelling and simulation.
- It forms a base distribution in many algorithms.

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4. Exponential Random Variable

4.1 Definition

An exponential random variable models time-related processes such as waiting time and decay-based behaviour.

4.2 Probability Density Function

For parameter $b > 0$, the PDF is defined as:

$$f_X(x) = \frac{1}{b} \exp\left(-\frac{x}{b}\right) u(x)$$

Where:

- $u(x)$ is the **unit step function**
- $u(x) = 0$ for $x < 0$
- $u(x) = 1$ for $x \geq 0$

Interpretation of Unit Step Function

The unit step function restricts the domain of the exponential random variable such that:

$$X \geq 0$$

This means the distribution exists only for non-negative values.

4.3 Cumulative Distribution Function

The CDF is given by:

$$F_X(x) = \left(1 - \exp\left(-\frac{x}{b}\right)\right) u(x)$$

4.4 Step-by-Step Derivation of CDF

The CDF is computed using integration of PDF.

Step 1: Start with definition

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$$

Step 2: Apply support restriction

Since exponential RV exists only for $t \geq 0$:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ \int_0^x \frac{1}{b} e^{-t/b} dt, & x \geq 0 \end{cases}$$

Step 3: Solve integral

$$\int_0^x \frac{1}{b} e^{-t/b} dt$$

Let:

$$I = \int e^{-t/b} dt$$

$$I = -be^{-t/b}$$

Applying limits:

$$F_X(x) = \left[-e^{-t/b}\right]_0^x$$

$$F_X(x) = 1 - e^{-x/b}$$

Step 4: Include unit step function

$$F_X(x) = (1 - e^{-x/b})u(x)$$

4.5 Graphical Behaviour

PDF Characteristics

- Maximum value occurs at $x = 0$
- PDF decreases exponentially as x increases
- Distribution is skewed to the right

CDF Characteristics

- Starts at zero
- Monotonically increases
- Approaches 1 asymptotically

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5. Example – Exponential Random Variable

Problem Statement

Let X be an exponential random variable with PDF:

$$f_X(x) = e^{-x}u(x)$$

Step 1: Identify Distribution Support

From unit step function:

$$X \geq 0$$

Step 2: Interpret Expression

The presence of $u(x)$ ensures:

- The PDF is zero for negative values.
- The exponential decay starts from $x = 0$.

Step 3: Observed Domain Representation

The lecture explains the domain as:

$$X \in (-\infty, \infty)$$

However, due to $u(x)$, the effective domain becomes:

$$X \in [0, \infty)$$

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6. Laplace Random Variable

The lecture lists the Laplace random variable as part of the covered distributions. The discussion includes:

- Example explanation
- Distribution usage in modelling
- Graph-based understanding

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7. Gamma Random Variable

The lecture introduces Gamma random variables and includes:

- Graph and special case analysis
- Example explanation
- Homework problem related to Gamma distribution

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8. Applications and Contextual Discussion

The lecture links continuous random variables with:

- Randomised algorithms
- Modelling processes
- Coding and inference mechanisms
- Data modelling and estimation
- Gaussian density estimation (in-class activity)

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9. Key Learning Observations

1. Continuous distributions are defined using PDF and CDF.

2. Uniform distribution models equal likelihood across intervals.
3. Exponential distribution models decay and waiting-time processes.
4. Unit step functions are used to restrict distribution domains.
5. Graphical interpretation helps understand behaviour of distributions.
6. These distributions are widely used in computing, modelling, and algorithmic design.

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10. Summary

This lecture develops foundational understanding of major continuous random variable distributions. The lecture systematically derives:

- Mathematical definitions
- PDF formulations
- CDF derivations
- Graphical interpretations
- Application contexts

The material forms a basis for probabilistic modelling and statistical analysis in computing systems.

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• Exam-ready bullet revision sheet
• Fully formatted LaTeX scribe
• Step-by-step solved derivation notes version
• Condensed cheat sheet

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